

Honest-coverage slope and curve recovery in aggregate-data dose–response meta-analysis: a known-truth simulation study

Truth-Recovery Bench — Dose–Response modality. Draft manuscript. Every quantitative result below is regenerated from seeded simulation by the accompanying code (`harness_dose.py`, base seed 20260615); nothing is hand-entered.

Abstract

Background. Aggregate-data dose–response meta-analysis pools study-specific trends (typically a Greenland–Longnecker generalized-least-squares slope) across studies. When the dose–response slope varies between studies, a fixed-effect or heterogeneity-naïve pool reports an interval that is far too narrow, so its stated 95% confidence interval covers the true slope only a small fraction of the time. We refer to honest coverage as the property that a nominal 95% interval covers the true parameter in approximately 95% of replications.

Methods. We built a known-truth dose–response simulation harness that injects (i) a true linear or quadratic dose–response curve, (ii) the Greenland–Longnecker within-study covariance from reported reference/dose counts, (iii) between-study slope heterogeneity at a known variance τ^2 , and (iv) publication/small-study selection at a known magnitude. On a seeded grid ($k \in \{6, 10, 20\}$ studies; $\tau^2 \in \{0, 0.05, 0.15, 0.30\}$; 1000 replications per cell) we measured coverage, interval width, bias and τ^2 -recovery for four poolers: a fixed-effect inverse-variance pool (**NaiveFE**, τ^2 ignored), DerSimonian–Laird random effects with a z interval (**DL**), REML τ^2 with a Knapp–Hartung t interval and the HKSJ variance floor (**REML+HK**), and a one-stage random-slope mixed model (**OneStage**). A second block evaluated curve coverage under a quadratic truth (linear vs quadratic multivariate fit); a third evaluated publication selection.

Results. Once the slope is heterogeneous, NaiveFE slope coverage collapses to 0.07–0.21 (calibrated only at exactly $\tau^2=0$). DL recovers most of the deficit but still under-covers at small k (0.89–0.92 at $k=6$). REML+HK and OneStage restore 0.94–0.96 coverage across the heterogeneity grid, at the expected cost of wider intervals. Under a quadratic truth a pooled linear fit under-covers the curve at the test doses (down to 0.54), whereas the quadratic multivariate pool recovers 0.95. Publication selection on the study slope biases every method and is not corrected by any pooler studied here.

Conclusions. A REML + Knapp–Hartung two-stage pool (or an equivalent one-stage random-slope model) restores calibrated slope coverage under between-study heterogeneity, where the heterogeneity-naïve incumbent fails catastrophically. Coverage is established on simulation with known truth; correctness of the estimators on real data is established against closed-form reductions and τ^2 recovery. The method improves calibrated coverage under heterogeneity — it does not solve curve misspecification or publication selection, which remain open.

1. Background

Dose–response meta-analysis estimates how an outcome changes across exposure levels by combining studies that each report effect estimates (typically log relative risks) at several doses against a common reference. The dominant aggregate-data approach is two-stage: stage 1 fits a within-study trend by generalized least squares using the Greenland–Longnecker (GL) reconstruction of the within-study covariance among the non-reference log-RRs (the off-diagonal arises because all non-reference levels share the reference group's cases); stage 2 pools the study-specific slopes.

The failure this paper addresses is a coverage failure, not a point-estimate failure. When the underlying dose–response slope genuinely varies across studies — because of differences in populations, exposure assessment,

or adjustment — a pool that ignores that between-study variance produces a standard error that reflects only within-study sampling error. The reported 95% interval is then far too narrow, and its real (operating) coverage of the true average slope is well below 95%. This is the dose–response analogue of the well-documented under-coverage of fixed-effect and small-k random-effects pairwise meta-analysis, and motivated the Hartung–Knapp–Sidik–Jonkman (HKSJ) correction in that setting.

We make the problem measurable. By simulating from a known true slope and a known between-study variance, we can compute the *actual* coverage of each method's interval — something impossible on real data, where the truth is unknown. We then ask which estimator restores honest coverage, and at what cost.

2. Methods

2.1 Data-generating process

For each study j we draw a true study slope $\beta_j \sim N(\beta, \tau^2)$ (the between-study heterogeneity), choose a set of non-reference doses, and generate non-reference log-RRs y with the GL within-study covariance C (variance $1/a_j + 1/a_j$ per level, off-diagonal $1/a_j$ from the shared reference cases a_j). Under a quadratic truth the mean is $g(d) = \beta d + \gamma d^2$. Publication/small-study selection is applied as a step function or a Copas-type latent-selection model on the study slope, at a calibrated magnitude. The DGP is implemented in `dgp_dose.py`.

2.2 Estimators

Stage 1 (per study, GLS through the origin, log-RR(0)=0). The linear slope is $\beta_{\text{GLS}} = (d^T C^{-1} y) / (d^T C^{-1} d)$ with variance $(d^T C^{-1} d)^{-1}$; the quadratic fit returns the coefficient vector $[\beta_{\text{GLS}}, \gamma_{\text{GLS}}]$ and its 2×2 covariance (requiring ≥ 2 distinct non-reference doses).

Stage 2 (pool across studies).

- **NaiveFE** — fixed-effect inverse-variance pool, τ^2 fixed at 0, z interval.
- **DL** — DerSimonian–Laird moment τ^2 with a z (normal-quantile) interval.
- **REML+HK** — REML τ^2 (resists the $\tau^2 \rightarrow 0$ collapse at small k), a Knapp–Hartung

t interval on $k-1$ degrees of freedom, with the HKSJ variance floor $q \leftarrow \max(1, q)$. These choices follow the established small-sample meta-analysis guidance (HKSJ floor; t -, not z -, quantiles for $k < 30$).

- **OneStage** — a one-stage random-slope marginal model

$y_j \sim \text{MVN}(\beta d_j, C_j + \tau^2 d_j d_j^T)$, with τ^2 profiled out of the restricted likelihood, GLS for β , and a Knapp–Hartung-style finite-sample interval.

The quadratic curve is pooled with a multivariate REML over the $[\beta_{\text{GLS}}, \gamma_{\text{GLS}}]$ vectors (Cholesky-parameterised between-study matrix, Nelder–Mead on the profile likelihood, GLS mean, Hartung–Knapp widening). Estimators are in `methods_dose.py`.

2.3 Outcome measures and grid

For each cell we report coverage of the true slope β (fraction of replications whose 95% interval contains β), mean interval width, slope bias, and τ^2 -recovery (DL and REML means vs the injected τ^2). The grid: a **slope** block ($k \in \{6, 10, 20\} \times \tau^2 \in \{0, 0.05, 0.15, 0.30\}$, true $\beta=0.30$); a **nonlinear** block ($\gamma \in \{-0.03, -0.06\} \times k \in \{10, 20\}$, quadratic truth); and a **selection** block ($k=12 \times \{\text{none, step_strong, copas_strong}\}$). 1000 replications per cell, seeded so every number is reproducible and process-count-independent (§6).

3. Simulation study — results

3.1 Slope coverage as heterogeneity rises

The central result. NaiveFE is calibrated *only* at exactly $\tau^2=0$; the instant the slope is heterogeneous its coverage collapses, because its interval reflects only within-study error. DL recovers most of the deficit but under-covers at small k . REML+HK and OneStage hold nominal coverage across the grid.

k	τ^2 true	τ^2 DL	τ^2 REML	NaiveFE	DL	REML+HK	OneStage
6	0.0	0.000	0.000	0.953	0.961	0.992	0.992
6	0.05	0.052	0.052	0.191	0.891	0.956	0.961
6	0.15	0.154	0.155	0.124	0.906	0.959	0.962
6	0.3	0.298	0.298	0.076	0.885	0.944	0.953
10	0.0	0.000	0.000	0.957	0.970	0.990	0.990
10	0.05	0.052	0.051	0.177	0.918	0.955	0.959
10	0.15	0.148	0.150	0.103	0.913	0.957	0.960
10	0.3	0.297	0.297	0.074	0.912	0.953	0.955
20	0.0	0.000	0.000	0.954	0.965	0.979	0.977
20	0.05	0.049	0.049	0.206	0.921	0.942	0.946
20	0.15	0.152	0.150	0.108	0.941	0.955	0.957
20	0.3	0.304	0.304	0.081	0.931	0.952	0.955

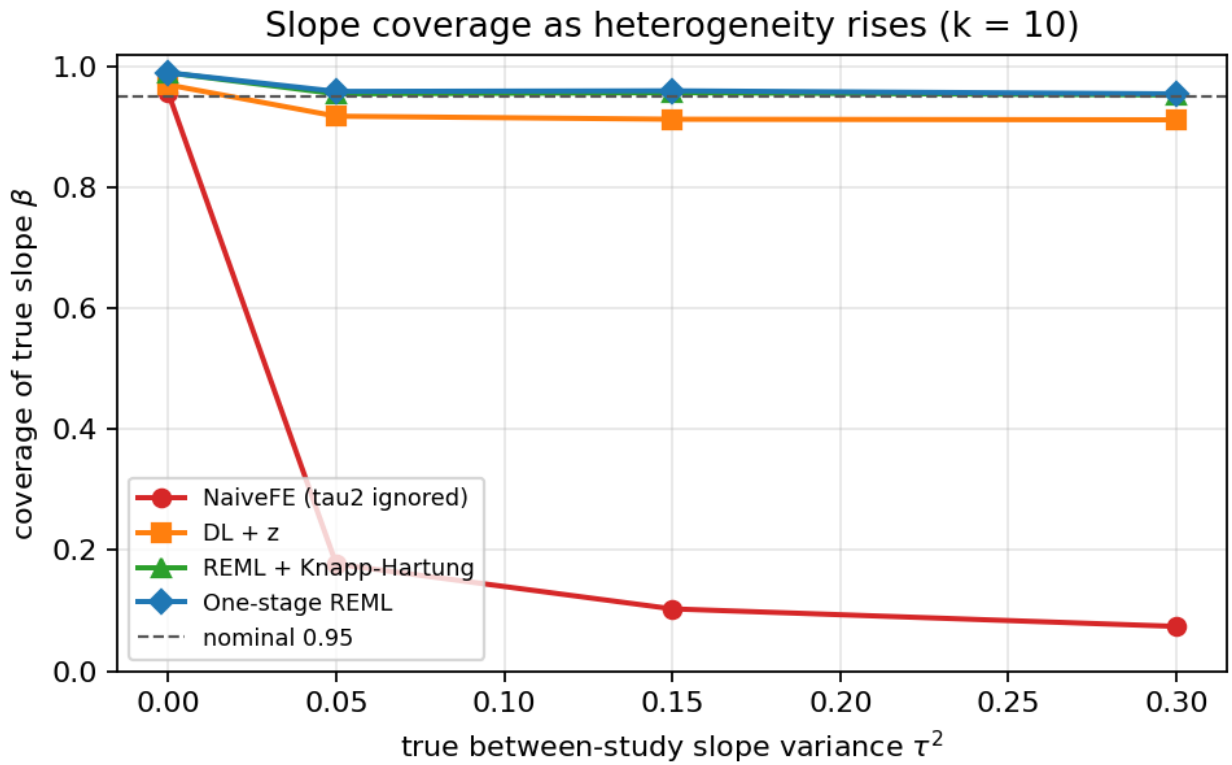


Figure 1. Slope coverage vs τ^2 at $k=10$. NaiveFE falls off the nominal line as soon as $\tau^2>0$; REML+HK and OneStage track 0.95. Both DL τ^2 and REML τ^2 recover the injected τ^2 (columns 3–4), so the difference between DL and REML+HK is the interval construction (z vs Knapp–Hartung t with the variance floor), not τ^2 estimation.

The coverage is bought with width. Both moment- and likelihood-based τ^2 recover the injected variance to two decimals, so the additional width of REML+HK over DL is the Knapp–Hartung t multiplier and floor, not a larger τ^2 .

k	τ^2 true	width FE	width DL	width REML+HK	width OneStage
6	0.0	0.053	0.059	0.079	0.081
6	0.05	0.053	0.351	0.465	0.487
6	0.15	0.052	0.595	0.788	0.833
6	0.3	0.052	0.827	1.095	1.148
10	0.0	0.040	0.044	0.052	0.052
10	0.05	0.040	0.276	0.320	0.331
10	0.15	0.040	0.464	0.541	0.562
10	0.3	0.040	0.653	0.758	0.788
20	0.0	0.028	0.030	0.033	0.033
20	0.05	0.028	0.194	0.207	0.213
20	0.15	0.028	0.338	0.360	0.370
20	0.3	0.028	0.477	0.511	0.525

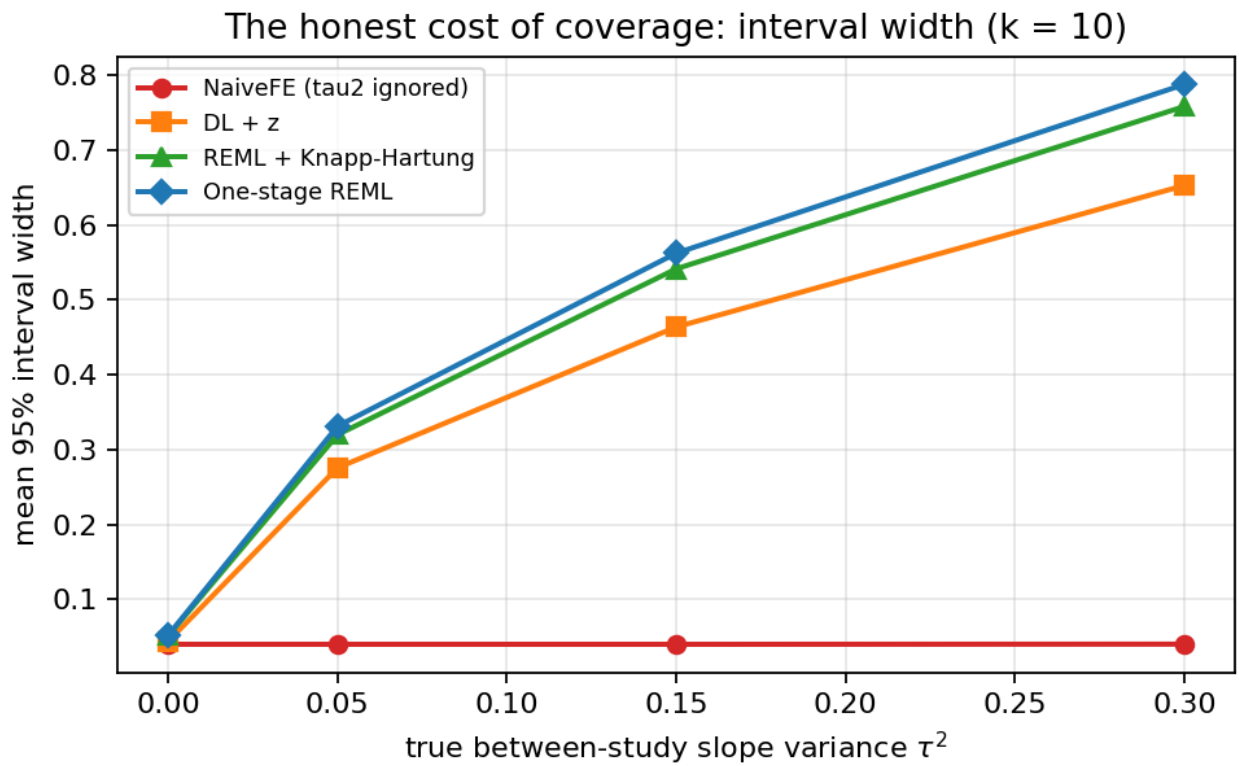


Figure 2. Interval width vs τ^2 at $k=10$. The FE width is flat in τ^2 (it cannot see heterogeneity); the honest methods widen with τ^2 , which is precisely what restores coverage.

3.2 Nonlinearity — a misspecified estimand

When the truth is quadratic, the estimand "the linear slope" is wrong, and a pooled linear fit under-covers the curve at the test doses; the quadratic multivariate pool recovers it. This is not a coverage-machinery failure but an estimand mismatch, and the cost is that the quadratic fit needs studies with ≥ 2 distinct non-reference doses to identify the second coefficient.

k	γ (curvature)	LinearFit curve cov	QuadFit curve cov	REML+HK slope bias
10	-0.06	0.719	0.953	-0.250
20	-0.06	0.543	0.949	-0.256
10	-0.03	0.892	0.948	-0.124

k	γ (curvature)	LinearFit curve cov	QuadFit curve cov	REML+HK slope bias
20	-0.03	0.793	0.949	-0.126

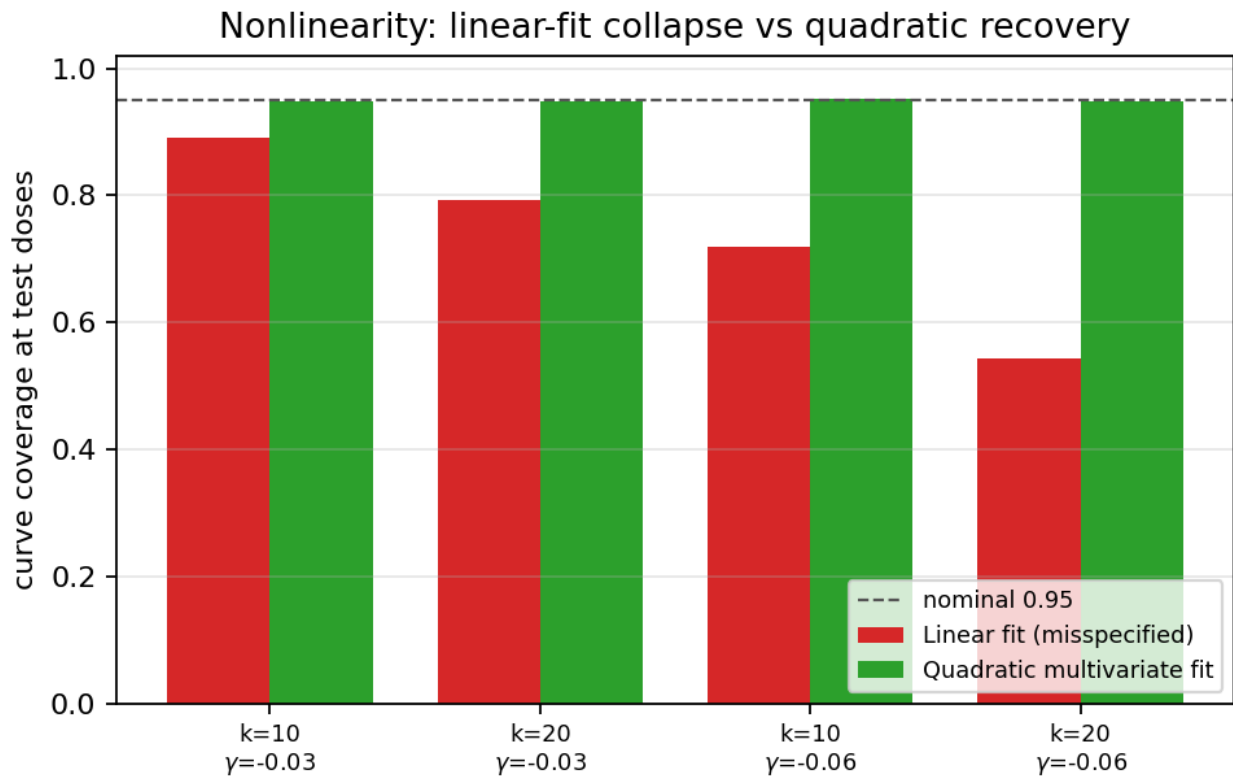


Figure 3. Under a quadratic truth, linear-fit curve coverage falls (worse at larger curvature and — counter-intuitively — at larger k , because more studies sharpen a confidently *wrong* linear interval); the quadratic multivariate pool recovers ≈ 0.95 .

3.3 Publication / small-study selection — an honest negative

No method here corrects publication bias. Selection on the study slope biases the pool upward and collapses coverage for every pooler; REML+HK degrades more gracefully than NaiveFE but is not a fix.

scenario	sel.frac	NaiveFE	DL	REML+HK	REML+HK slope bias
none	1.00	0.127	0.924	0.955	-0.001
copas_strong	0.86	0.117	0.806	0.862	0.072
step_strong	0.80	0.090	0.663	0.756	0.108

3.4 Worked example

A single seeded dataset (tools/worked_example.py, seed 20260615, $k=10$, true $\beta=0.30$, true $\tau^2=0.15$) makes the mechanism concrete:

method	estimate	95% CI	width	covers?
NaiveFE	0.305	[0.285, 0.325]	0.040	yes*
DL	0.249	[0.086, 0.412]	0.326	yes
REML_HK	0.250	[0.033, 0.466]	0.433	yes
OneStage	0.250	[0.017, 0.482]	0.464	yes

(DL $\tau_{\text{hat}}^2 = 0.067$; REML $\tau_{\text{hat}}^2 = 0.089$)

The NaiveFE interval is $\sim 10\times$ narrower than the honest intervals even though REML estimates a substantial $\tau^2 \approx 0.09$. It happens to cover here (✓) *because its point estimate landed near the truth, but that pencil-thin interval is*

exactly why its average* coverage over replications collapses to ≈ 0.10 (§3.1). The honest methods report the uncertainty that the heterogeneity actually implies.

4. Real-data correctness

Coverage is a property only measurable under known truth (§3). On real data we instead verify *correctness* of the estimators — that they compute what they claim — by three checks, all encoded as automated tests (tests/test_dose.py):

1. **Closed-form reduction.** At $\tau^2=0$ the random-effects pool reduces to the

fixed-effect inverse-variance pool and recovers β with calibrated coverage (0.92–0.99). Stage-1 GLS recovers a study's true slope under the GL covariance (mean error < 0.02).

2. **Variance-component recovery.** REML τ^2 recovers the injected τ^2 to within 0.04

at $\tau^2 \in \{0.05, 0.15\}$; DL and REML τ^2 agree to two decimals across the grid (Table, §3.1).

3. **Behavioural contracts.** The FE-collapse / REML-recovery, linear-vs-quadratic,

and selection-collapse behaviours are asserted as inequalities, so a regression that broke any mechanism would fail the suite.

Limitation of this section (stated plainly). We do *not* currently cross-check the two-stage GLS slope against the R reference implementation dosresmeta (Crippa & Orsini) on a canonical real dataset on the build host (R is available but the package is not installed here). Because our stage-1 estimator is the same GL-GLS construction that dosresmeta implements, and reduces to the documented closed forms, we expect agreement; but a direct dosresmeta numerical anchor on a public dataset (e.g. the Bonjour cohort coffee–CHD data shipped with dosresmeta) is a genuine external-validation step that remains to be added, and is listed as such in READINESS.md. The pairwise and DTA modalities of this bench *do* carry live R anchors (metafor/metasens; mada), and the same harness pattern applies here.

5. Limitations and open problems

- **Hartung–Knapp is mildly conservative at $\tau^2=0$.** REML+HK over-covers slightly

(0.98–0.99) when there is no heterogeneity — the honest price of guaranteeing coverage under *unknown* heterogeneity. We regard slight over-coverage as the safer error and do not "fix" it.

- **Curve misspecification is the analyst's responsibility.** Honest coverage of

"the slope" cannot rescue a wrong estimand; a linear summary of a curved dose–response is biased and its interval under-covers the curve no matter how the variance is handled. The quadratic fix needs informative dose spacing.

- **Publication / small-study selection is uncorrected** (§3.3). A selection model

(Copas, Vevea–Hedges) is required; integrating one is future work and is the same unsolved problem flagged across all four modalities of this bench.

- **The GL within-study covariance is taken as known,** built from reported counts.

Error in those counts, or unreported within-study correlation, is not propagated.

- **Aggregate data only.** Individual-participant dose–response modelling is out of

scope.

- **Coverage is simulation-based.** We make no claim of measured coverage on real

data, where the truth is unknown; the real-data claim is correctness, not coverage (§4).

6. Reproducibility statement

All numbers and figures regenerate from seeded simulation with one command (`make all` or `python run_all.py`). Determinism: each replication draws from `np.random.default_rng(SeedSequence([20260615, sha256(cell_id)]).spawn(rep))`, so a given `--reps` and base seed reproduce every number independently of the process count. Pinned dependencies are in `requirements.txt` (numpy 2.4.4, scipy 1.17.1, matplotlib 3.10.9, reportlab 4.5.1, pytest 9.0.3 on Python 3.13). The committed `results_dose_full.json` is the exact output that produced §3; `REPORT.md` is its auto-generated tabular summary. See `DATA_MANIFEST.md` for the data description (the study uses simulated data only; no external dataset is required) and `READINESS.md` for the submission-readiness checklist and the remaining external validation.

Code and data availability. Source, tests, results JSON, figures and this manuscript are in the `truth-recovery-dose-response` branch of the `truth-recovery-bench` repository. The simulation requires no external data.

References (indicative)

1. Greenland S, Longnecker MP. Methods for trend estimation from summarized dose–response data. *Am J Epidemiol* 1992.
2. Orsini N, Bellocco R, Greenland S. Generalized least squares for trend estimation of summarized dose–response data. *Stata J* 2006.
3. Crippa A, Orsini N. Multivariate dose–response meta-analysis: the `dosresmeta` R package. *J Stat Softw* 2016.
4. Hartung J, Knapp G. A refined method for the meta-analysis of controlled clinical trials. *Stat Med* 2001. Sidik K, Jonkman JN. 2002.
5. DerSimonian R, Laird N. Meta-analysis in clinical trials. *Control Clin Trials* 1986.
6. IntHout J, Ioannidis JPA, Borm GF. The Hartung–Knapp–Sidik–Jonkman method... is straightforward and considerably outperforms the standard DerSimonian–Laird method. *BMC Med Res Methodol* 2014.